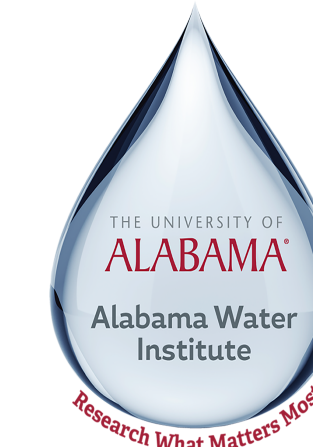




The NextGen Research Datastream: A Numerical Prediction System for Accelerating Research to Operations in Hydrology

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NextGen Research Datastream

<https://github.com/CIROH-UA/ngen-datastream>

The NRDS is

- a numerical prediction system for accelerating the flow of research into operations
- a system for rapid deployment and refinement of new methods in an operational-like testbed
- A community resource of public data, tooling, and infrastructure that meets calls for reproducibility

The datastream design concept has already enabled

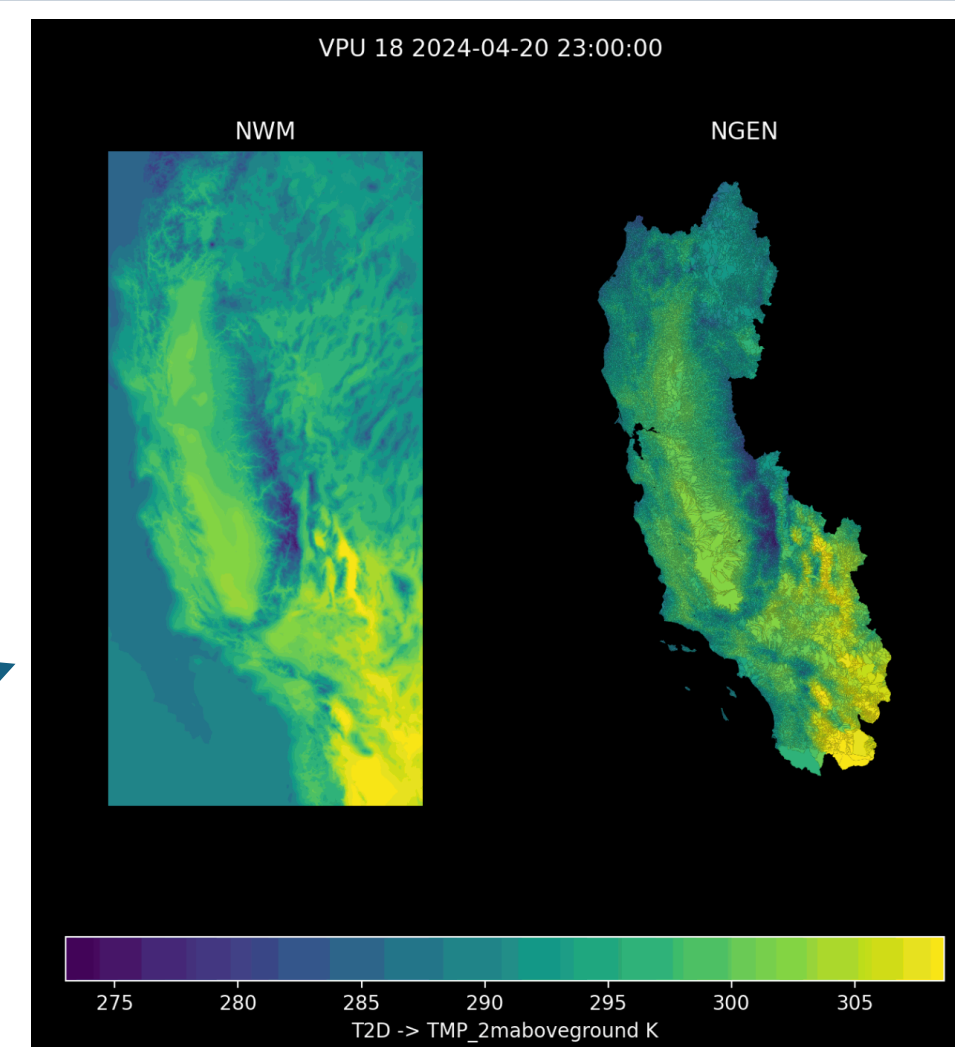
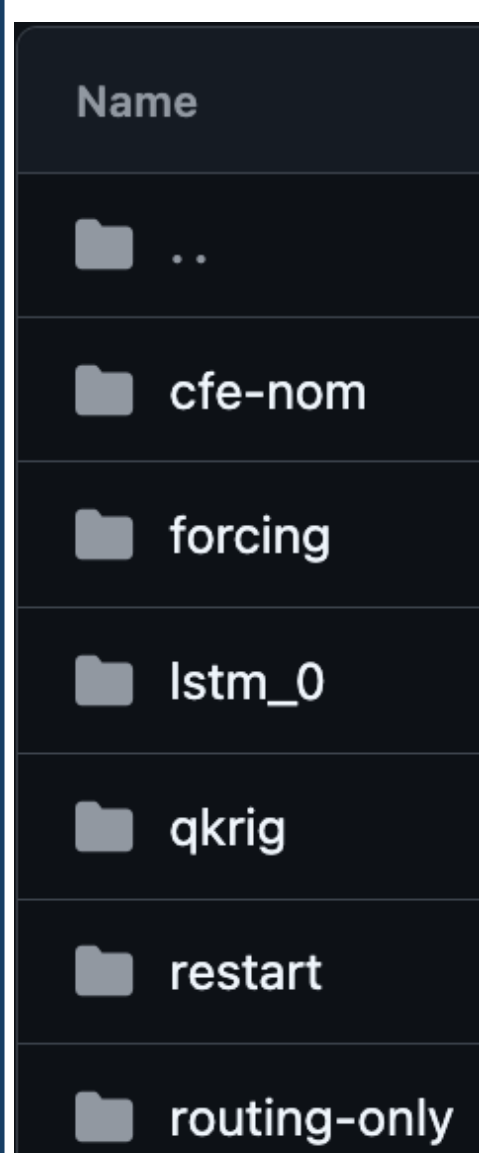
- 1000+ daily simulations
- CONUS-scale simulations
- Concurrency between model formulations



NRDS Blog Post



Deployed Datastreams



Go to the repo's status and metadata markdown for a timeline of NRDS deployments



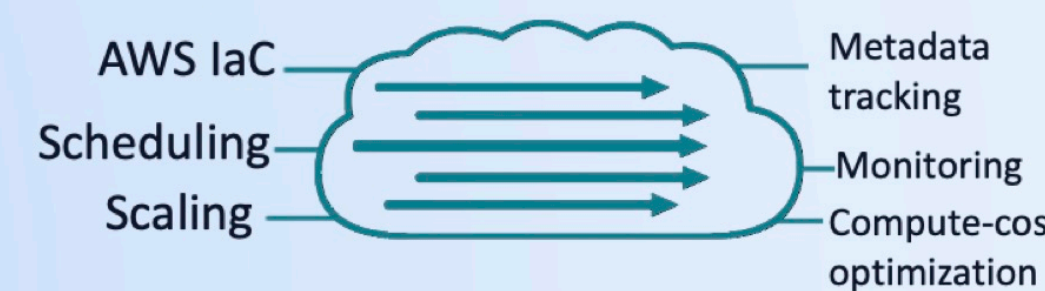
NextGen Research DataStream [NRDS]

- Daily NextGen simulations across CONUS VPUs.
- Public outputs at datastream.ciroh.org; visualizer at nrds.ciroh.org.



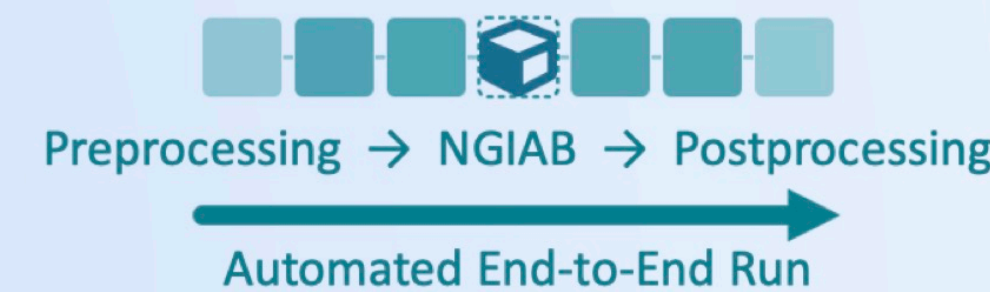
Datastream

- Infrastructure-as-code on AWS
- Scales DataStreamCLI to scheduled, parallel runs with metadata tracking and compute-cost optimization.



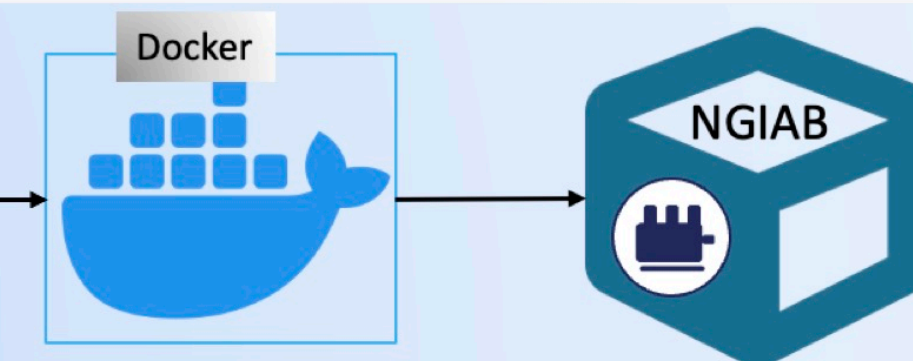
DataStreamCLI

- Stand-alone Command Line Interface (CLI)
- Automates the full pipeline: hydrofabric subset, forcings, NGIAB, and output handling — reproducibly



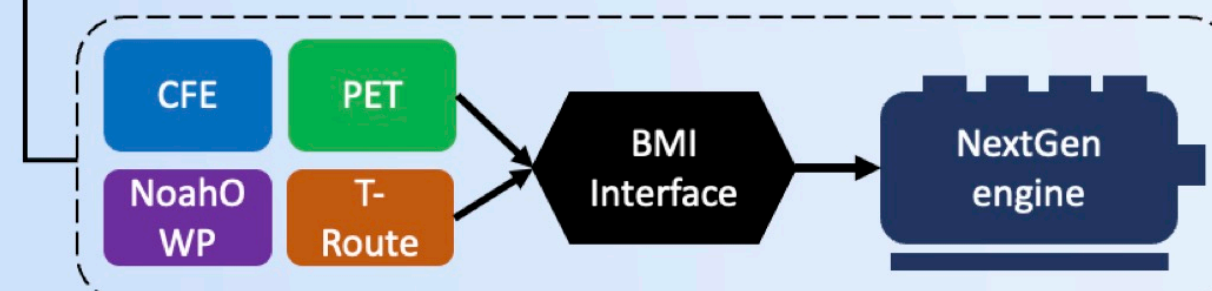
NextGen In A Box [NGIAB]

- Containerized NextGen
- Docker images that package the framework



NextGen Framework

- Modeling engine
- Model/module library

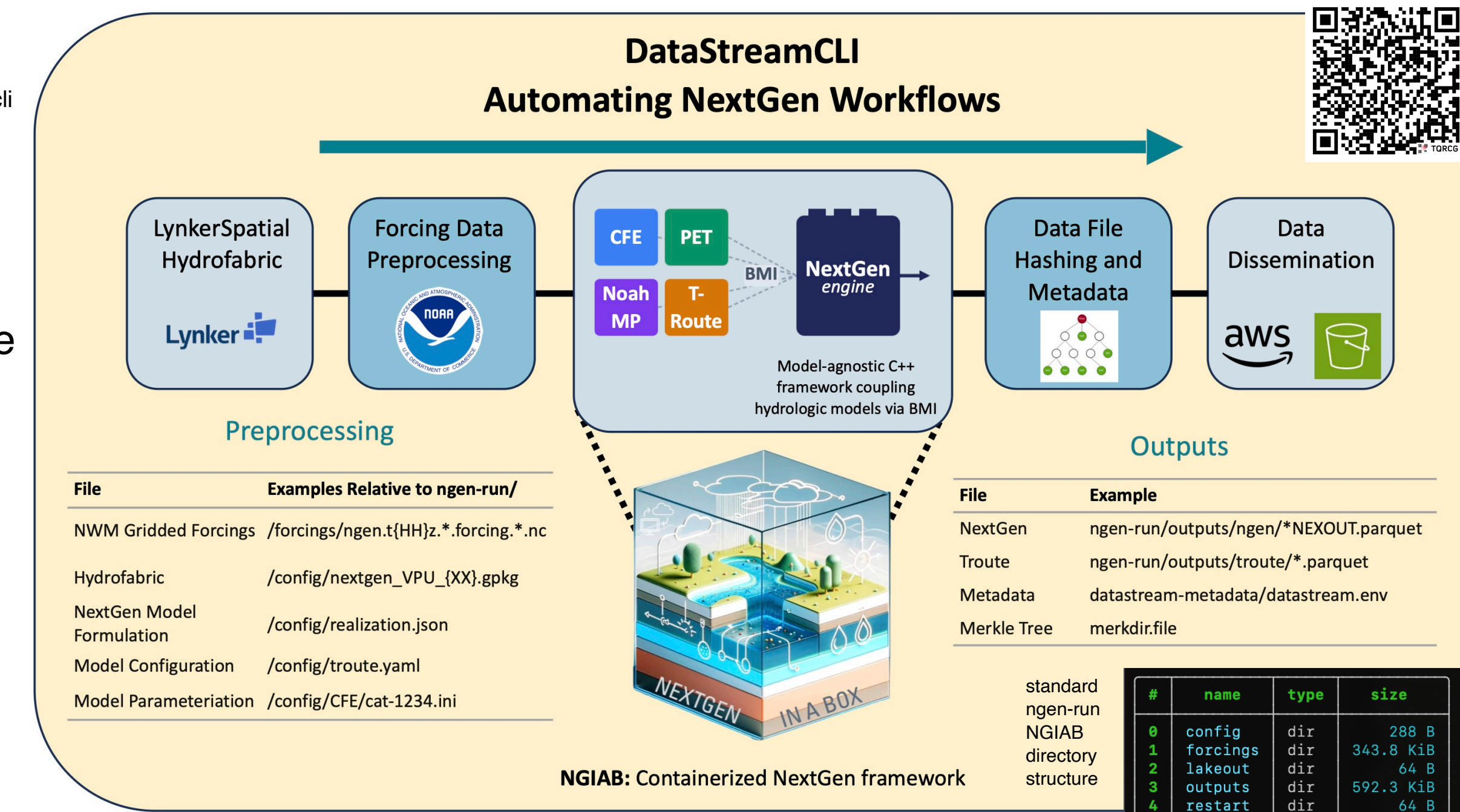


DataStreamCLI

<https://github.com/CIROH-UA/datastreamcli>

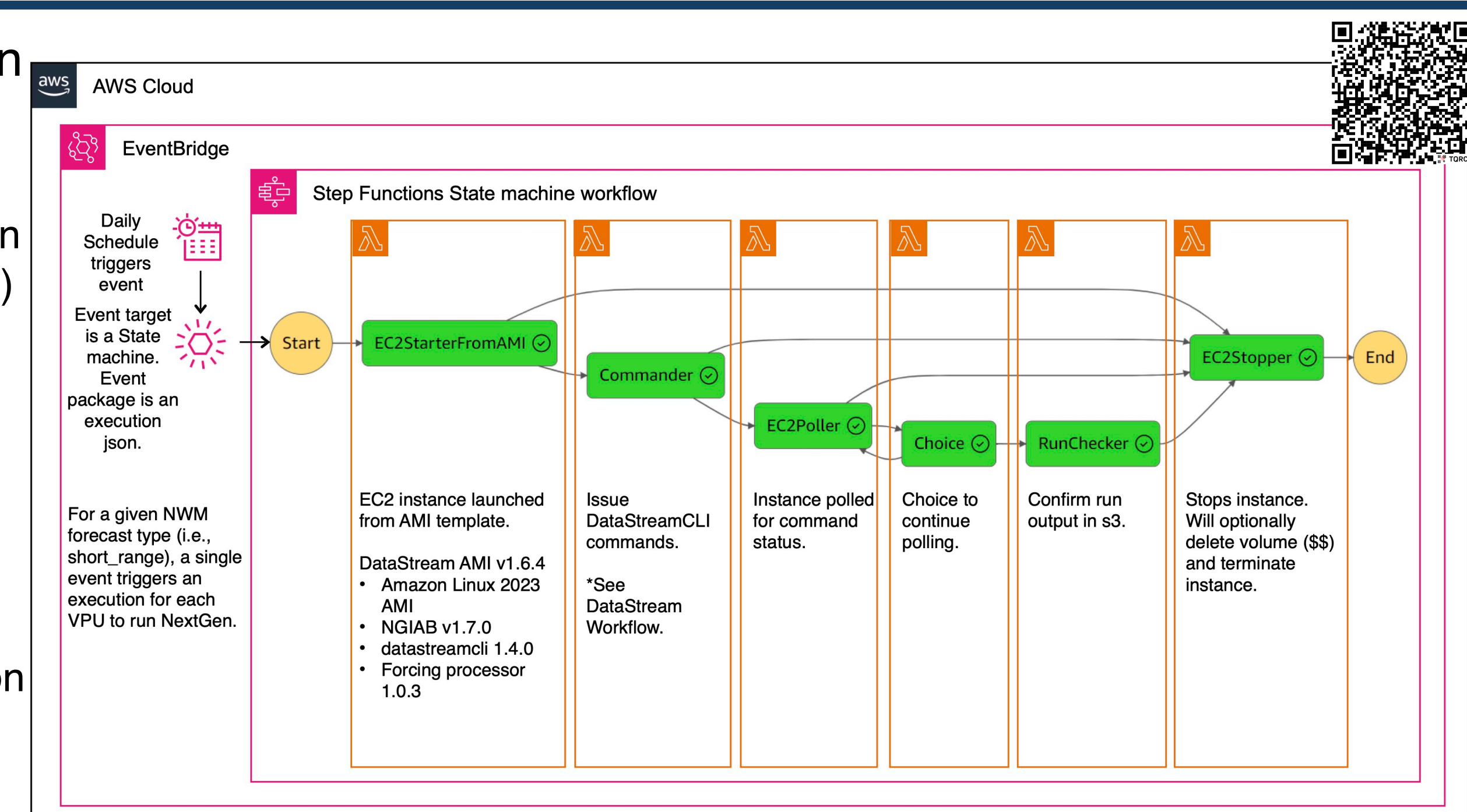
DataStreamCLI is the layer at which the datastream abstraction becomes a reproducible computation.

This workflow tool is open and updated alongside the NRDS, maintaining reproducibility for the community,



AWS Infra Design

The NRDS is conducted and scheduled in Amazon Web Services (AWS) cloud. The design below is responsible for ensuring the outputs from single NRDS simulation. Initial status results show >99.9% simulation completion making the NRDS reliable as a prediction system.



Future Work

- Continue to accept new datastreams as there is currently large demand from the community!
- Deploy dHBV as a datastream
- Integrate community contributed parameters
- Academic article
- Continue writing documentation for community contributions and governance

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Community Contributions

Ways to contribute

- New datastream (NextGen formulation)
- Updated Hydrofabric
- Calibrated parameter contributions
- Updated weights for LSTM
- Updated model configuration files (BMI)
- Any update that improves the accuracy of outputs or reduces compute cost

Go to the repo's issues to initiate a community contribution!



See the completed qkrig contribution!



See the ongoing dHBV contribution!

